

AI Powered Phishing Attack Detection Using Machine Learning

Project Title:

AI Powered Phishing Attack Detection Using Machine Learning

1. Abstract

Phishing attacks are one of the most common cybersecurity threats where attackers create fake websites or URLs to steal sensitive information. This project presents an AI-based phishing detection system that uses Machine Learning techniques to identify whether a website URL is safe or potentially malicious.

The system uses a phishing URL dataset, extracts important features from URLs, and applies a Random Forest Machine Learning algorithm for classification. A Flask-based web application is developed to provide an interactive interface where users can enter a URL and receive a prediction result.

2. Introduction

With the rapid growth of internet usage, phishing attacks have increased significantly. Attackers use fake websites that look similar to legitimate websites to deceive users. Traditional detection methods are often unable to identify new phishing techniques. Machine Learning provides an effective approach by learning patterns from previous phishing data and predicting suspicious URLs. This project focuses on developing an automated phishing detection system using Artificial Intelligence.

3. Objectives

To detect phishing websites using Machine Learning.

To analyze URL-based features for identifying malicious websites.

To develop a user-friendly web interface for phishing detection.

To improve cybersecurity awareness and protection against phishing attacks.

4. Tools and Technologies Used

Programming Language:

Python

Libraries:

Pandas

NumPy

Scikit-learn

Joblib

Flask

Frontend:

HTML

CSS

Machine Learning Algorithm:

Random Forest Classifier

Development Environment:

Visual Studio Code

Windows 10

5. Methodology

The project follows these steps:

Dataset Collection

A phishing website dataset is collected containing different URL features.

Data Processing

The dataset is loaded and analyzed using Python libraries.

Model Training

A Random Forest Classifier is trained using phishing URL data.

Feature Extraction

Important URL characteristics are extracted for prediction.

Web Application Development

A Flask-based interface is created for user interaction.

Prediction

The trained model classifies URLs as Safe or Phishing.

6. System Architecture

User enters URL

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URL Feature Extraction

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Machine Learning Model

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Random Forest Classifier

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Prediction Result

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Safe Website / Phishing Website

7. Results

The developed system successfully detects suspicious URLs.

Model Accuracy:

96.74%

The system provides:

Safe Website detection

Phishing Website detection

Web-based user interface

8. Conclusion

The AI Powered Phishing Detection System successfully demonstrates the use of Machine Learning in cybersecurity. The system can analyze URL patterns and classify websites as safe or phishing. This project helps in reducing the risk of phishing attacks and improving online security.

9. Future Scope

Real-time phishing detection.

Browser extension integration.

Deep Learning based detection.

Threat intelligence API integration.

Advanced URL analysis techniques.